CHAPTER 6

Pairing Correlations and Superfluid Nuclei*

6.1 Introduction and Experimental Survey

In Chapter 5 we looked for a wave function describing the ground state of the nucleus. Restricting ourselves to a product ansatz and minimizing the total energy of the system has led us to the Hartree–Fock method. As we have seen, the solution of the corresponding equation yields a transformation from a given single-particle basis to a new, better one. The variational principle is equivalent to the requirement that there are no matrix elements between the ground state and the most simple excitations, the particle-hole excitations. Therefore, the Hartree–Fock method partially takes into account the particle-hole part of the interaction, that is, the long-range part of the force, as we have seen in Chapter 4. Before we turn to the excitations caused by these correlations, we want to consider the short-range part of the force which causes particle–particle correlations (cf. Chap. 4). It will turn out that this can be done formally very similarly to the particle-hole part of the force by introducing generalized product wave functions consisting of "quasi-particles."

Of course we usually have to take into account both correlations at the same time. This will be done in the next chapter within the framework of Hartree–Fock–Bogoliubov theory. Here we restrict ourselves to pure

* We are glad to see that the original version of this chapter (in the book by Baumgärtner and Schuck) [BS 68a] has partly been adopted by other authors [EG 70]. We have also incorporated some of their changes.

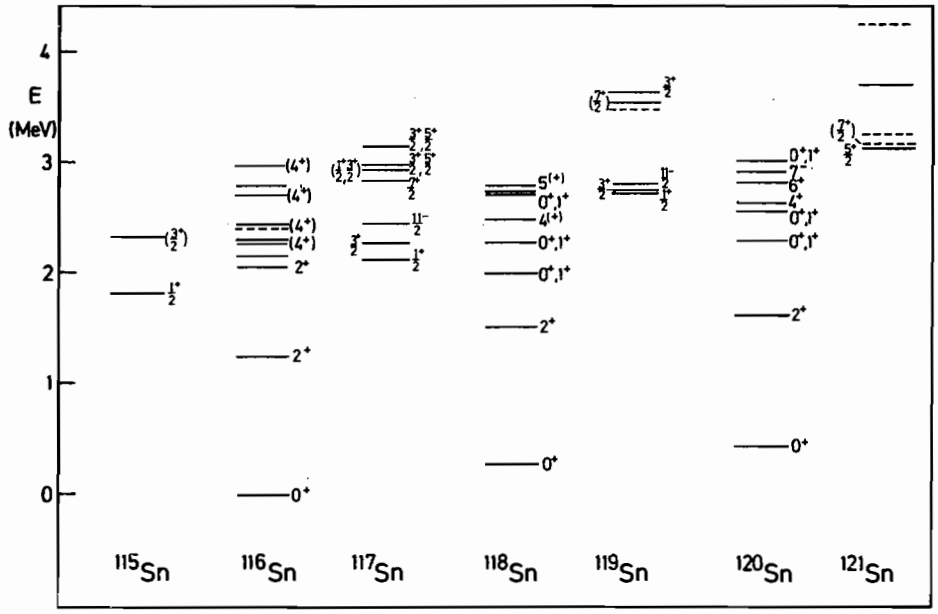


Figure 6.1. Excitation spectra of the $_{50}$Sn isotopes.

short-range (particle–particle or pairing) correlations. This allows us to understand many important effects which cannot be explained within a pure Hartree–Fock picture.

Let us briefly summarize the essential experimental facts which lead to the idea of pairing correlations:

(i) *The energy gap.* The spectra of deformed nuclei show a characteristic difference between even and odd nucleon number. Even–even nuclei have only few (collective) levels up to 1.5 MeV excitation energy. They can be nicely interpreted as rotational and vibrational bands. The situation is very different for even–odd nuclei, which have many collective and single-particle states in the same energy interval. Figure 6.1 shows the spectra of some tin isotopes as an example.

(ii) *The level density.* If we assume that there are only a few nucleons in a shell of given j, many states can be constructed which are all energetically degenerate corresponding to the various possibilities of coupling angular momentum. The number of states per energy unit can easily be estimated, and it is found that in the low-lying excitation region the level density exceeds that found experimentally by roughly a factor of two.

(iii) *Odd–even effect.* The total binding energy of an odd–even nucleus is found to be smaller than the arithmetic mean of the binding energies of the two neighboring even–even nuclei. Therefore, we

have the following relation for the masses of neighboring nuclei:

$$M_{(A \text{ odd})} > \frac{M_{A-1} + M_{A+1}}{2}.$$

This is called the odd–even effect.

(iv) The *moments of inertia* of deformed nuclei can be measured from the level structure of rotational bands. Calculations based on the pure single-particle model (see Sec. 3.4.2) deviate by a factor of two from the experimental values. If pairing is included, theory and experiment are in much better agreement.

(v) *Deformations*. If, in the pure shell model, we calculate the density distribution of the nucleons as a function of the nuclear mass number, we find that there is a steady transition from spherically symmetric shapes for closed shell nuclei to strong deformations for nuclei with half-filled shells. Nuclei whose mass numbers do not deviate very much from the closed shell configuration, however, stay at least in their ground state, spherically symmetric. Filling more nucleons into the shell, one enters a region in which nuclei undergo rapid changes in deformation, reaching its maximum value in the middle of the shell.

(vi) *Low-lying 2^+ states*. We find in even nuclei, in the neighborhood of closed shell nuclei, a low-lying level with angular momentum 2 and positive parity (Fig. 6.1). These levels can be interpreted neither as rotations nor as single-particle excitations. In fact, they are vibrational in character (see Chap. 8), having a strong interplay with pairing correlations.

To understand all these phenomena we have to take into account the correlations due to the short-range part of the nucleon–nucleon interaction (see Chap. 4). As indicated in Chapter 2, even–even nuclei have ground state spin $I_0=0$, and the spin of odd–even nuclei is determined by the angular momentum of the odd nucleon. These observations led Göppert–Mayer [Ma 50] to a very early calculation showing that for *short-ranged, attractive, two-nucleon forces*, the coupling of two nucleons in a shell of given j to a $(I=0)$-*pair* is energetically favored over all other possible couplings. This can be understood very easily by neglecting the spin for a moment and by considering the density distribution of states $|lm\rangle$ and $|l-m\rangle$, as shown schematically in Fig. 6.2 (see also the discussion on this point in Section 4.4.7). From these density distributions, it is clear that the spatial overlap of two-nucleon densities is maximal if the two nucleons have the same $|m|$. For a short-range force the configuration in which the two nucleons orbit the nucleus with equal $|m|$ but in an opposite sense is therefore energetically very much favored (the case where they turn around in equal sense is obviously unfavored by the Pauli principle; see, however, the case of angular alignment in Chap. 3). Here, orbiting in opposite sense means coupling to $I=0$. The angular part of a pair coupled to $I=0$, $M=0$

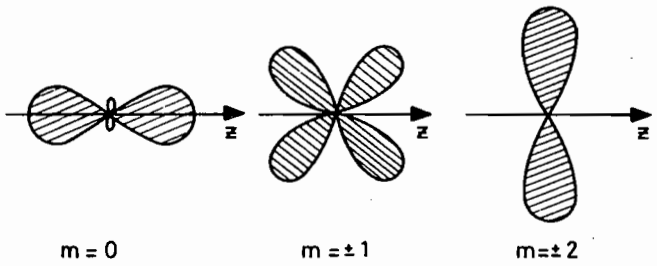


Figure 6.2. Density distributions for a nucleon in a given magnetic substate.

is, to within some factors, given by (4.132):

$$\langle \theta_1\varphi_1, \theta_2\varphi_2 \,|\, nljjI=0, M=0\rangle \propto \sum_{m=-l}^{l} (-1)^m Y_{lm}(\theta_1\varphi_1)\, Y_{l-m}(\theta_2\varphi_2)$$

$$= \frac{2l+1}{4\pi} P_l(\cos\theta_{12}), \tag{6.1}$$

where θ_{12} is the angle between the two nucleons. From (6.1), we see that the pair wave function depends only on the relative angle θ_{12}. For a fixed θ_{12}, the pair wave function is therefore uniformly distributed around a (spherical) nucleus. Since the radial functions of the two nucleons are the same, the position of the pair in space cannot, in general, be given in more detail, the radial dependence being very much a question of the nodal structure of the nucleon wave functions. As we have already seen in Fig. 4.9, $P_l(\cos\theta_{12})$ is peaked at $\theta_{12}=0$ with a width of $1/l$. Therefore, the two nucleons have a tendency to stay close in angle, which is necessary for their *spatial overlap* to be *maximal* imposed by the short range of the force. On the other hand, the two nucleons cannot get too close because this is forbidden by the uncertainty principle [it can be easily verified that the kinetic energy would be infinite for the pair wave function $\psi(\mathbf{r}_1, \mathbf{r}_2) \sim \delta(\mathbf{r}_1 - \mathbf{r}_2)$]. We can therefore say that the nucleons want to stay as close as possible according to their short-range interaction, but that the pair has to make a compromise with its kinetic energy, which would like to keep the nucleons apart.

If there is *more than one pair* in the j-shell, the formation of $(I=0)$-pairs is somewhat hindered by the Pauli principle. Some m-states necessary to achieve the coupling (6.1) of one pair are already occupied by another pair. This mutual interaction of pairs gives rise to certain special features which we will study later on but which do not destroy the pair-state character of a j-shell. On the other hand, some typical features, like transition probabilities, are proportional to the number of pairs we have in a given j-shell. If we take account of the symmetry between particles and holes, that is, of the fact that we have particle–particle as well as hole–hole pairs, we see that the effect of collectivity is largest for half-filled j-shells. On the other hand, for closed shell nuclei, no formation of pairs like (6.1) is possible.

For nuclei between closed shells, the energetically most favored configuration will be the one in which all nucleons (except the last one in the case

of odd–even nuclei) are paired off. To excite even–even nuclei, therefore, we have to break at least one pair, whereas for odd–even nuclei an excitation can be achieved by putting the odd nucleon in a very low energy level. Since the binding energy of a pair is of the order 1–2 MeV, the different energy spacing between the ground and first excited state in even–even and odd–even nuclei is thus qualitatively explained (see Fig. 6.1).

Other features mentioned above can also be understood with the pairing model. Because of the energy gap in the low-lying part of the spectrum of even–even nuclei, the level density agrees much better with the experimental one. Also the odd–even effect becomes obvious. The existence of $(I=0)$-pairs favors—as we have seen (6.1)—a spherical nuclear shape (no direction is preferred). Nuclei in the neighborhood of closed shells will therefore still have spherical symmetry, since the influence of the pairing force overcomes the tendency to deform. Further away from the closed shells we will have the opposite situation. However, this depends very sensitively on the strength of the pairing force versus the long-range $(p\text{-}h)$ force; nevertheless, in this way the rather sudden change from sphericity to deformation can be understood.

Closely connected is the existence of *low-lying* 2^+ *levels* for open shell even–even nuclei. Nuclei in the neighborhood of closed shells that are still spherical can easily be excited to shape vibrations around their spherical equilibrium position, since the restoring force, which is the difference between pairing and deformation effects, is rather small. The nucleus will therefore become deformed into an ellipsoid and vibrate about its spherical shape with a low frequency (quadrupole oscillations, 2^+-levels).

The diagonalization of the pair interaction cannot be interpreted as a contribution to the average static potential of the Hartree–Fock type. It is a completely new effect which gives rise to the so-called pairing potential. It is analogous to superconductivity in metals. This is the reason why Bohr, Mottelson, and Pines [BMP 58] and Belyaev [Be 59] successfully applied the methods of the theory of superconductivity by Barden, Cooper, and Schrieffer [BCS 57] to nuclei.

6.2 The Seniority Scheme

As we said in the introduction, pairing correlations are due to the short-range part of the nucleon–nucleon interaction, and this interaction is most effective between $(I=0)$-coupled pairs. In Section 4.4.7 we studied this problem to some length; to investigate the pairing phenomena in more detail we reconsider the model force derived in (4.140) and consider N particles in a single $(2j+1)$-fold degenerate j-shell interacting through this pairing force. If we place this j-shell at zero energy, the corresponding

Hamiltonian is of the form*:

$$H = -G \sum_{m,\, m' > 0} a_m^+ a_{-m}^+ a_{-m} a_m = -G S_+ S_- , \tag{6.2}$$

with

$$S_+ = \sum_{m>0} a_m^+ a_{-m}^+ ; \qquad S_- = (S_+)^+ . \tag{6.3}$$

We introduced the operators of Eq. (6.3) because it turns out that they can be considered as the raising and lowering operators of a fictitious angular momentum, which we will call (as in the case of the Lipkin model; see Chap. 5) *quasi-spin*, which has nothing to do with real spin, but its introduction greatly facilitates the solution of (6.2) [Ke 61]. For this purpose we introduce the following three quantities for each substate m $(m>0)$:

$$\begin{aligned} s_+^{(m)} &= a_m^+ a_{-m}^+ ; \\ s_-^{(m)} &= a_{-m} a_m ; \\ s_0^{(m)} &= \tfrac{1}{2}(a_m^+ a_m + a_{-m}^+ a_{-m} - 1). \end{aligned} \tag{6.4}$$

Using the commutation rules of the operators a_m and a_m^+, we can derive the corresponding commutation rules:

$$\begin{aligned} \left[s_+^{(m)}, s_-^{(m)} \right] &= 2 s_0^{(m)} ; \\ \left[s_0^{(m)}, s_+^{(m)} \right] &= s_+^{(m)} ; \\ \left[s_0^{(m)}, s_-^{(m)} \right] &= - s_-^{(m)} . \end{aligned} \tag{6.5}$$

Therefore, we see that the triad of operators $s_+^{(m)}, s_-^{(m)}$, $s_0^{(m)}$ has the commutation properties of angular momentum operators [Ed 57]. The $s_+^{(m)}, s_-^{(m)}$ are, respectively, the raising and lowering operators analogous to the angular momentum operators j_+ and j_-. Similarly, $s_0^{(m)}$ corresponds to the z-component of the angular momentum j_0. We thus see the reason why we call $\mathbf{s}^{(m)}$ the quasi-spin operator corresponding to the level m.

Furthermore, we can see from the definition of $s_0^{(m)}$ that it has eigenvalues $\pm \frac{1}{2}$ depending on whether the pair $(m, -m)$ is full or empty. The vector $\mathbf{s}^{(m)}$, therefore, has a spin of $\frac{1}{2}$ angular momentum for 0, or two particles in the j-level. [If only one particle is present, all the components of $\mathbf{s}^{(m)}$ are zero, so that $\mathbf{s}^{(m)}$ has spin zero in this subspace.]

The total spin vector $\mathbf{S}$ is defined by

$$\mathbf{S} = \sum_{m>0} \mathbf{s}^{(m)} . \tag{6.6}$$

This, of course, is also an angular momentum, and the pairing Hamiltonian (6.2) is conveniently written in the form:

$$H = -G(\mathbf{S} \cdot \mathbf{S} - S_0^2 + S_0). \tag{6.7}$$

*Note that we use BCS-phases (4.138) in the following.

We note also that

$$S_0 = \frac{1}{2} \sum_{m>0} (a_m^+ a_m + a_{-m}^+ a_{-m} - 1) = \frac{1}{2}(\hat{N} - \Omega), \tag{6.8}$$

where $\Omega = j + \frac{1}{2}$ is the number of pairs. Thus the eigenvalues of S_0 are integers or half integers, depending on whether S_0 is an integer or a half integer. Furthermore, it follows from the properties of angular momentum operators that

$$S \geqslant |S_0| = \tfrac{1}{2}|N - \Omega|. \tag{6.9}$$

Since, according to Eq. (6.6), the maximum value of S is $S = \Omega/2$, we see that S can take on all values $\Omega/2$, $(\Omega/2) - 1$, $(\Omega/2) - 2, \ldots, |(\Omega/2) - (N/2)|$. The states will then be labeled by S and S_0, and the energy eigenvalues of H (6.6) are given by

$$E(S) = -G\left\{S \cdot (S+1) - \tfrac{1}{4}(N-\Omega)^2 + \tfrac{1}{2}(N-\Omega)\right\}. \tag{6.10}$$

As an alternative quantum number to the total quasi-spin S, we can introduce the so-called *seniority quantum number* s [Ra 43] given by the relation

$$S = \tfrac{1}{2}(\Omega - s), \tag{6.11}$$

where

$$\begin{aligned} s &= 0, 2, 4, \ldots N \quad \text{for } N \text{ even}, \\ s &= 1, 3, 5, \ldots N \quad \text{for } N \text{ odd}. \end{aligned}$$

The total energy (6.10) as a function of seniority is then given by

$$\begin{aligned} E(N, s) &= -\frac{G}{4}(N - s)(2\Omega - s - N + 2) \\ &= -\frac{G}{4}\left(s^2 - 2s(\Omega + 1) + 2N(\Omega + 1) - N^2\right). \end{aligned} \tag{6.12}$$

The characteristic feature of the force is that the energy is labeled only by the seniority and is degenerate in all other quantum numbers except N. Although the absolute value of the energy depends strongly on the number of particles, the spectrum does not. Since the total quasi-spin changes by units of one, we have [from (6.10)] for the energy difference of two neighboring levels

$$E(S-1) - E(S) = 2G \cdot S. \tag{6.13}$$

The binding energy increases for a given N with S; the ground state is therefore obtained when all quasi-spins are aligned, that is, $S = \frac{1}{2}\Omega$ or $s = 0$. For an even system, the first excited state has $s = 2$; and from (6.12) or (6.13) we have, for the excitation energy of the first excited state,

$$E(N, s=2) - E(N, s=0) = G \cdot \Omega. \tag{6.14}$$

The excitation energies are thus independent of the number of particles in the j-shell.

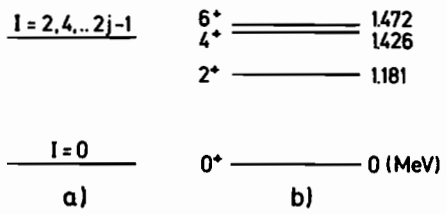


Figure 6.3. (a) Two-particle spectrum of a pure pairing force. (b) Spectrum of ^{210}Po (2p in the $1h9/2$ shell).

If we have only two particles in the j-shell, the seniority can take on only the values $s=0,2$. The two (identical) nucleons can couple to $I=0,2,4,\ldots,2j-1$. Only the state with $I=0$ is affected—that is, lowered—by the pairing interaction (6.2). The other states are unaffected and therefore degenerate. Thus we see that the ground state ($s=0$) has $I=0$ and the excited states ($s=2$) have $I=2,4,6,\ldots$, that is, in the ground state the two nucleons are "paired," whereas in the excited states the pair is "broken." This fact is represented schematically in Fig. 6.3, where we also show for comparison the low-lying spectrum of ^{210}Po (two protons in a $1h9/2$ shell). In the general case, the binding energy of the ground state as a function of N can be written as:

$$E(N,s=N)-E(N,s=0)=-E(N,s=0)=G\Omega\frac{N}{2}\left(1-\frac{N-2}{2\Omega}\right). \quad (6.15)$$

We see that for cases in which the particle number is much smaller than the degeneracy of the shell ($N\ll\Omega$), the binding increases linearly with the number ($n=N/2$) of pairs, that is, the total energy is just the binding energy of one pair (6.14) multiplied by the number of pairs. If we draw the ground state energies as a function of n, we obtain a "harmonic spectrum" (Fig. 6.4). This is the *pair vibrational spectrum*, which is found, for example, in the lead region and which we will consider in more detail in Section 8.3.5. All the states of the spectrum in Fig. 6.4 have maximal quasi-spin ($s=0$), that is, they are *ground states*. We speak then of the $s=0$ band; there are, of course, $s=2$ bands, etc. It has been found that two-particle transfers between two states of a given band are enhanced with respect to those between two states belonging to different bands. If the number of pairs increases, the mutual disturbance of $I=0$ pairs due to the Pauli principle (see introduction) becomes important and the spectrum (6.15) becomes anharmonic.

We see that the excitation energy of the *first excited state* (6.14) is equal to the binding energy of one pair (6.15). Alternatively, we can say that in

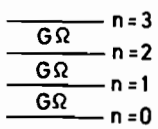


Figure 6.4. Harmonic spectrum of pair vibrations.

the first excited state ($s=2$) two nucleons are unpaired. (The possibility of creating the first excited state by putting the bound pair into a higher level is excluded in our model, since there is only one level.) This classification scheme straightforwardly generalizes to the statement that a state of seniority s has s unpaired particles (s even and odd).

The ground state is apparently never degenerate, since there is only one way to make a total quasi-spin of $S=\Omega/2$ out of Ω spins $\frac{1}{2}$. There are $\Omega-1$ ways for the $s=2$ state:

$$t_2=\Omega-1=\binom{\Omega}{1}-\binom{\Omega}{0}. \tag{6.16}$$

This can be easily checked by considering a definite example, say $\Omega=4$; the $s=2$ state can then be represented as: $\overset{1}{\rightarrow}\ \overset{2}{\rightarrow}\ \overset{3}{\rightarrow}\ \overset{a}{\leftarrow}$. There are $4-1=3$ possibilities of grouping a with one of the quasi-spins, namely $a1, a2, a3$.

It can be verified that for an arbitrary, even s the degree of degeneracy is given by [BS 68a]:

$$t_s=\binom{\Omega}{s/2}-\binom{\Omega}{s/2-1}, \qquad s\leqslant N. \tag{6.17}$$

The *wave function* corresponding to Eq. (6.2) can also be obtained easily using the quasi-spin formalism. For $N=0$ (a completely empty shell) and $N=2\Omega$ (a completely full shell), the third component of each of the quasi-spins is $-\frac{1}{2}$ or $+\frac{1}{2}$, respectively. Therefore, the vacuum is the state $S=\frac{1}{2}\Omega$, $S_0=-\frac{1}{2}\Omega$,

$$|-\rangle=|\tfrac{1}{2}\Omega, -\tfrac{1}{2}\Omega\rangle, \tag{6.18}$$

and the full shell is the state $S=\frac{1}{2}\Omega$; $S_0=\frac{1}{2}\Omega : |\Omega/2, \Omega/2\rangle$. If we represent a general quasi-spin state by $|SS_0\rangle$, where $S_0=\frac{1}{2}(N-\Omega)$, the *ground states* ($S=\Omega/2$) for different even N can be written as

$$|\tfrac{1}{2}\Omega, S_0\rangle=|\tfrac{1}{2}\Omega, \frac{N-\Omega}{2}\rangle \propto S_+^{N/2}|-\rangle. \tag{6.19}$$

The application of the raising operator S_+ (6.3) increases the number of particles by two. The ground states are eigenstates of S^2, S_0, and H and are just the products of $n=N/2$ pair states. Therefore, we also say that the ground state is a *pair condensate*.

For *odd N-systems*, the last particle cannot be paired and one of the quasi-spins will necessarily be zero. The largest possible value of S will therefore be $\frac{1}{2}(\Omega-1)$ and the ground states are given by:

$$|\frac{\Omega-1}{2}, S_0\rangle \propto S_+^{\frac{1}{2}(N-1)} a_m^+ |-\rangle. \tag{6.20}$$

Since m can take on the 2Ω different values $-j\leqslant m\leqslant j$, the ground state of an N-odd system is 2Ω-fold degenerate (in contrast to the nondegenerate ground state of even systems). This is the explanation of why the level density for odd systems close to the ground state is high, whereas for even systems there is always an energy gap.

As we have seen, the seniority 2 states ($S=\frac{1}{2}\Omega-1$) are the *first excited states* for even N systems. They can also be easily constructed within the quasi-spin formalism.

We first consider the case $N=2$ and construct eigenstates with good angular momentum by applying the operators

$$A^+_{IM}=\frac{1}{\sqrt{2}}\sum_{m_1m_2} C^{j\ j\ I}_{m_1m_2M}a^+_{m_1}a^+_{m_2} \tag{6.21}$$

on the vacuum

$$|IM\rangle=A^+_{IM}|-\rangle. \tag{6.22}$$

Since $A^+_{00}\propto S_+$ [we use BCS phases, Eq. (4.138)], they are eigenstates of the quasi-spin with $S=\Omega/2$ for $I=0$ and with $S=\frac{1}{2}\Omega-1$ for $I\neq 0$. To see this, we calculate

$$\mathbf{S}^2|IM\rangle=\left[\mathbf{S}^2,A^+_{IM}\right]|-\rangle+\tfrac{1}{2}\Omega(\tfrac{1}{2}\Omega+1)|IM\rangle. \tag{6.23}$$

The first term on the r.h.s. is:

$$\begin{aligned}\left[\mathbf{S}^2,A^+_{IM}\right]|-\rangle&=\left[S_+S_-,A^+_{IM}\right]|-\rangle+\left[S_0^2-S_0,A^+_{IM}\right]|-\rangle\\&=S_+\left[S_-,A^+_{IM}\right]|-\rangle+\left[S_0^2-S_0,A^+_{IM}\right]|-\rangle.\end{aligned} \tag{6.24}$$

Since S_- is proportional to A_{00}, it follows from the orthogonality relations for the Clebsch–Gordon coefficients [Ed 57] that for $I\neq 0$,

$$\left[S_-,A^+_{IM}\right]|-\rangle=0. \tag{6.25}$$

Using (6.8), we finally find for $I\neq 0$:

$$\begin{aligned}\mathbf{S}^2|IM\rangle&=-\Omega|IM\rangle+\frac{\Omega}{2}\left(\frac{\Omega}{2}+1\right)|IM\rangle\\&=\left(\frac{\Omega}{2}-1\right)\frac{\Omega}{2}|IM\rangle.\end{aligned} \tag{6.26}$$

Therefore, the two-particle states (6.22) are, for $I\neq 0$, eigenstates of $\mathbf{S}^2$ with $S=(\Omega/2)-1$ or $s=2$, that is, the two particles are unpaired in the $I\neq 0$ states.

Since the operator S_+ cannot change S and only serves to add two particles, it is easy to construct the first excited states for an even N system with $N>2$:

$$|\tfrac{1}{2}\Omega-1,S_0\rangle\propto S_+^{\frac{1}{2}(N-2)}A^+_{IM}|-\rangle. \tag{6.27}$$

Since S_+ is a spherical tensor operator of rank zero, these states have the angular momentum quantum numbers I, M and seniority two for $I\neq 0$.

The states (6.27) are, of course, eigenstates of the Hamiltonian (6.2) with the energy (6.12), and we see that it does not depend on I, M. As we have previously stated, the higher excited states are characterized by the number of unpaired particles, that is, the seniority number s. Such states can be constructed using multiple application of the A^+_{IM} operators, but the

orthogonality to lower states as well as the construction of states with good angular momentum is then nontrivial.

In many cases, the explicit construction of the wave function, however, is not needed (e.g., many matrix elements can be obtained using angular momentum coupling techniques), which is one of the advantages of the quasi-spin formalism.

Let us calculate, for example, the *matrix elements of the quadrupole operator*:

$$Q_{20} = \sum_{m=-j}^{j} q_m a_m^+ a_m = \sum_{m>0} q_m s_0^{(m)}$$

$$q_m = \langle jm | r^2 Y_{20} | jm \rangle. \tag{6.28}$$

From (6.28) we see that $\hat{Q}_{20}$ is a vector in quasi-spin space. This fact implies that the dependence of any quadrupole matrix element on $S_0 = \frac{1}{2}(N-\Omega)$, that is, on the particle number, is given by the Wigner–Eckart theorem [Ed 57]:

$$\langle SS_0 | Q_{20} | S'S_0 \rangle = (-)^{S-S_0} \begin{pmatrix} S & 1 & S' \\ -S_0 & 0 & S_0 \end{pmatrix} \langle S \| Q_2 \| S' \rangle. \tag{6.29}$$

Since quadrupole transitions go from the ground state with seniority zero to an excited state with $s=2$ [selection rule given by (6.29)], we get for the quadrupole transition probability

$$|\langle s=2 | Q_{20} | s=0 \rangle|^2 \propto \begin{pmatrix} \Omega/2 & 1 & \Omega/2-1 \\ -S_0 & 0 & S_0 \end{pmatrix}^2 \propto \left(\frac{\Omega}{2} - S_0\right)\left(\frac{\Omega}{2} + S_0\right)$$

$$= \frac{N}{2}\left(\Omega - \frac{N}{2}\right). \tag{6.30}$$

From (6.30) we see the *collective effect* for transition probabilities which we mentioned in the introduction: It is proportional to the product of the number of particle pairs $N/2$ and the number of hole pairs $(\Omega - N/2)$. The collective effect is therefore most pronounced in the middle of closed shells and due to the aforementioned condensation of pairs in the ground state. This is in agreement with the experimental finding that the quadrupole transition probabilities for nuclei in the middle of closed shells are enhanced with respect to their pure shell model values.

The model we presented in this section is indeed very simple and in reality things will be much more complicated. We will have to consider, for instance, more levels and more realistic forces. Nevertheless, we wish to stress the point that many features of the seniority model remain valid in more realistic cases, and in a certain sense the mathematical and physical transparency of this model reflects reality better than the quite abstract formalism of quasi-particles, which we introduce in the next section, used to handle more complex situations.

6.3 The BCS Model

6.3.1 The Wave Function

The applicability of the seniority model is not limited to j^N configurations. If there exist several almost degenerate j-shells above the closed core, the model can be generalized [Ta 71, 76, LA 74, Lo 75]. Far away from closed shells, however, where nuclei are deformed and the levels more or less uniformly separated, the seniority model breaks down completely.

However, the results of the seniority model show that pairing correlations are very important in nuclei with open shells, and there is no reason to believe that this is no longer true still further from closed shells. In the following we therefore present a method which allows us to treat all the nucleons in the nucleus and which can easily be generalized for different types of interactions. This method no longer provides an exact solution of the eigenvalue problem, but, like the Hartree–Fock method, it can be derived from a variational principle. Therefore, it will be important to make the right ansatz for the ground state wave function. In analogy to Bardeen, Cooper, and Schrieffer [BCS 57], who determined the ground state of a superconductor, we try to represent the wave function for even–even nuclei in the following way.

$$|\mathrm{BCS}\rangle = \prod_{k>0} (u_k + v_k a_k^+ a_{\bar{k}}^+)|-\rangle, \tag{6.31}$$

where u_k and v_k represent variational parameters. The product runs only over half the configuration space, as indicated by $k>0$. For each state $k>0$ there exists a "conjugate" state $\bar{k}<0$ and the states $\{k, \bar{k}\}$ generate the whole single-particle space.

The v_k^2 and u_k^2 represent the probability that a certain pair state $(k, \bar{k})$ is or is not occupied, which has to be determined in such a way that the corresponding energy has a minimum. They are not, however, independent, as the norm of the state (6.31) requires

$$|u_k|^2 + |v_k|^2 = 1. \tag{6.32}$$

Since the BCS wave function is only fixed to within a phase factor, it is always possible to choose the coefficients u_k real and positive. In principle, the phase of v_k has to be determined by the variation of the energy expectation value. It can be shown, however (see Sec. 6.3.4), that for certain reasonable assumptions for the interaction, real positive values v_k yield the deepest energy. The ansatz (6.31) contains the coefficients u_k and v_k only for $k>0$. In the following it is sometimes convenient to also use coefficients $u_{\bar{k}} := u_k$ and $v_{\bar{k}} := -v_k$ for the values $\bar{k}<0$.

In many cases, especially if the Hamiltonian is invariant under time reversal, the conjugate state can be chosen as the time-reversed state [Me 61 Ch. XV]:

$$|\bar{k}\rangle = T|k\rangle. \tag{6.33}$$

An example is a spherical basis [in BCS phases, Eq. (4.138)]:

$$|k\rangle = |nljm\rangle, \qquad |\bar{k}\rangle = |nlj - m\rangle, \qquad m > 0. \tag{6.34}$$

The following formulae, however, apply also for cases of Hamiltonians without time reversal symmetry (see Sect. 7.7).

At this point in the discussion, the *variational ansatz* (6.31) for the wave function seems rather arbitrary. It will become clear in the next chapter that it emerges quite naturally by a slight modification of the Hartree–Fock ansatz. Let us simply make the observation that the particles appear in mutually conjugate pairs similar to the ground state (6.19) of the seniority model, but with the important difference that $|\mathrm{BCS}\rangle$ is a superposition of different numbers of pairs, that is, (6.31) no longer has a sharp particle number. This is actually a great disadvantage of (6.31) in nuclear physics.

The product can be written as

$$|\mathrm{BCS}\rangle \propto |-\rangle + \sum_{k>0} \frac{v_k}{u_k} a_k^+ a_{\bar{k}}^+ |-\rangle + \frac{1}{2} \sum_{kk'>0} \frac{v_k v_{k'}}{u_k u_{k'}} a_k^+ a_{\bar{k}}^+ a_{k'}^+ a_{\bar{k}'}^+ |-\rangle + \cdots. \tag{6.35}$$

In solid state physics, where $N \simeq 10^{23}$, the violation of particle number has no influence on any physical quantity. In nuclei, however, the violation of the invariance corresponding to the particle number in many cases gives rise to serious errors. One then has to use improved methods to deal with such problems (PBCS, FBCS; see Sec. 11.4.3).

To give an impression of the flexibility of the ansatz (6.31), we rewrite it in a different way: It can be expressed by a *generalized pair creation operator* (6.3)

$$A^+ = \sum_{k>0} \frac{v_k}{u_k} a_k^+ a_{\bar{k}}^+ \tag{6.36}$$

as

$$|\mathrm{BCS}\rangle \propto \exp(A^+)|-\rangle = \sum_{\nu=0}^{\infty} \frac{1}{\nu!} (A^+)^\nu |-\rangle. \tag{6.37}$$

The component having the particle number N is therefore $(A^+)^{N/2}$. This corresponds to the seniority zero state ($s=0$) of Eq. (6.19); in fact, this component is exactly the ground state wave function of the seniority model, since in this case all v_k's and u_k's are equal, as we shall see later on. In that sense, the BCS-ansatz contains the $s=0$ state of independent pairs, that is, it is a condensate of bound pairs (boson-like entities; see Chap. 9), which are in the same quantum state ($I=0$, $T=1$). In infinite matter this leads to a Bose–Einstein condensation of the pairs (superconductivity). In finite systems like nuclei there exists no real phase transition. Nevertheless, it can come quite close to it and in any case (6.37) should be a good ansatz.

6.3.2 The BCS Equations

We assume that a many-body system is described by the Hamiltonian

$$H=\sum_{k_1k_2\gtrless 0} t_{k_1k_2}a_{k_1}^{+}a_{k_2}+\frac{1}{4}\sum_{\substack{k_1k_2k_3k_4\\ \lessgtr 0}} \bar{v}_{k_1k_2k_3k_4}a_{k_1}^{+}a_{k_2}^{+}a_{k_4}a_{k_3}. \tag{6.38}$$

The parameters u and v of the trial wave function (6.31) are determined by variation of the energy. However, this variation is *restricted* by the subsidiary condition that the expectation value of the particle number has the desired value N

$$\langle \mathrm{BCS}|\hat{N}|\mathrm{BCS}\rangle = 2\sum_{k>0} v_k^2 = N. \tag{6.39}$$

This can be achieved by adding the term $-\lambda\hat{N}$ to the variational Hamiltonian

$$H' = H-\lambda\hat{N}. \tag{6.40}$$

The *Lagrange multiplier* λ is fixed by the condition (6.39). It is called the *chemical potential* or the *Fermi energy* because it represents the increase of the energy $E=\langle \mathrm{BCS}|H|\mathrm{BCS}\rangle$ for a change in the particle number

$$\lambda = \frac{dE}{dN}. \tag{6.41}$$

To see this, we use the fact that the expectation value of H' is a minimum with respect to an arbitrary variation of the BCS wave function (6.31). One special variation is a change of the parameter λ. Therefore, we get [see also Eq. (3.81)]

$$\frac{d}{d\lambda'}\left\{\langle \mathrm{BCS}(\lambda')|H|\mathrm{BCS}(\lambda')\rangle-\lambda\langle \mathrm{BCS}(\lambda')|\hat{N}|\mathrm{BCS}(\lambda')\rangle\right\}_{\lambda=\lambda'}=0 \tag{6.42}$$

or

$$\frac{dE}{d\lambda}=\lambda\frac{dN}{d\lambda}. \tag{6.43}$$

In the following we will always use H' instead of H. For the calculation of actual energy, however, we have to remember that we have to add the term λN at the end.

Another interesting quantity in this connection is the particle–number uncertainty

$$(\Delta N)^2 := \langle \mathrm{BCS}|\hat{N}^2|\mathrm{BCS}\rangle - N^2 = 4\sum_{k>0} u_k^2v_k^2, \tag{6.44}$$

which we will discuss later on.

From (6.31) and (6.38), we gain for the BCS expectation value of H':

$$\langle \mathrm{BCS}|H'|\mathrm{BCS}\rangle = \sum_{k\gtrless 0}\left\{(t_{kk}-\lambda)v_k^2+\frac{1}{2}\sum_{k'\gtrless 0}\bar{v}_{kk'kk'}v_k^2v_{k'}^2\right\} + \sum_{kk'>0}\bar{v}_{k\bar{k}k'\bar{k}'}u_kv_ku_{k'}v_{k'}. \tag{6.45}$$

Since the BCS wave function is completely determined by the parameters v_k and the condition (6.32), the variation

$$\delta\langle \mathrm{BCS}|H'|\mathrm{BCS}\rangle = 0 \tag{6.46}$$

yields

$$\left(\frac{\partial}{\partial v_k} + \frac{\partial u_k}{\partial v_k}\frac{\partial}{\partial u_k}\right)\langle \mathrm{BCS}|H'|\mathrm{BCS}\rangle = 0. \tag{6.47}$$

After differentiating, we finally obtain the set of BCS equations

$$2\tilde{\epsilon}_k u_k v_k + \Delta_k\left(v_k^2 - u_k^2\right) = 0, \qquad k > 0, \tag{6.48}$$

with

$$\tilde{\epsilon}_k = \frac{1}{2}\left(t_{kk} + t_{\bar{k}\bar{k}} + \sum_{k' \gtrless 0} (\bar{v}_{kk'kk'} + \bar{v}_{\bar{k}k'\bar{k}k'})v_{k'}^2\right) - \lambda \tag{6.49}$$

and the gap parameters (for real matrix elements)

$$\Delta_k = -\sum_{k'>0} \bar{v}_{k\bar{k}k'\bar{k}'} u_{k'} v_{k'}. \tag{6.50}$$

For fixed values $\tilde{\epsilon}_k$ and Δ_k, (6.32) and (6.48) yield two quadratic equations for u_k^2 and v_k^2, respectively, having the solutions

$$\begin{aligned} v_k^2 &= \frac{1}{2}\left[1 \pm \frac{\tilde{\epsilon}_k}{\sqrt{\tilde{\epsilon}_k^2 + \Delta_k^2}}\right] \\ u_k^2 &= \frac{1}{2}\left[1 \pm \frac{\tilde{\epsilon}_k}{\sqrt{\tilde{\epsilon}_k^2 + \Delta_k^2}}\right]. \end{aligned} \tag{6.51}$$

In the case of no interaction one has $\Delta = 0$ and $v_k^2 = 1$, $u_k^2 = 0$ for occupied orbits ($\tilde{\epsilon} < 0$). The only possible solutions of (6.51) are therefore:

$$\begin{aligned} v_k^2 &= \frac{1}{2}\left[1 - \frac{\tilde{\epsilon}_k}{\sqrt{\tilde{\epsilon}_k^2 + \Delta_k^2}}\right], \\ u_k^2 &= \frac{1}{2}\left[1 + \frac{\tilde{\epsilon}_k}{\sqrt{\tilde{\epsilon}_k^2 + \Delta_k^2}}\right]. \end{aligned} \tag{6.52}$$

Thus the variational principle (6.46) yields the set of equations (6.49), (6.50), and (6.52). Together with the *particle-number condition*

$$2\sum_{k>0} v_k^2 = N \tag{6.53}$$

they allow calculation of the BCS parameters u_k, v_k. In general, these equations are nonlinear and have to be solved by iteration.

For discussion of the properties of these equations it is often useful to

insert (6.52) into (6.50) and obtain the so-called *gap equation*:

$$\Delta_k = -\frac{1}{2}\sum_{k'>0} \bar{v}_{k\bar{k}k'\bar{k}'} \cdot \frac{\Delta_{k'}}{\sqrt{\tilde{\epsilon}_{k'}^2 + \Delta_{k'}^2}} \,. \tag{6.54}$$

6.3.3 The Special Case of a Pure Pairing Force

As we have seen in Section 6.2, the pure pairing force provides a very simple and powerful model for the description of pairing properties in nuclei. It is therefore widely used in the BCS description of nuclei. In the following, we therefore present the most important formulae of this theory for this special case.

The Hamiltonian here has the form,

$$H = \sum_{k>0} \epsilon_k (a_k^+ a_k + a_{\bar{k}}^+ a_{\bar{k}}) - G \sum_{kk'>0} a_k^+ a_{\bar{k}}^+ a_{\bar{k}'} a_{k'} \,. \tag{6.55}$$

The expectation value of H' [Eq. (6.45)] is

$$\langle \mathrm{BCS}|H'|\mathrm{BCS}\rangle = 2\sum_{k>0}\left(\tilde{\epsilon}_k v_k^2 + \frac{1}{2} G v_k^4\right) - \frac{\Delta^2}{G} \,. \tag{6.56}$$

In this case the gap parameter Δ does not depend on k:

$$\Delta = G \sum_{k>0} u_k v_k \,. \tag{6.57}$$

$\tilde{\epsilon}_k$ is given by

$$\tilde{\epsilon}_k = \epsilon_k - \lambda - G \cdot v_k^2 \,. \tag{6.58}$$

The term $G v_k^4$ in (6.56) and $G v_k^2$ in (6.58) is often neglected, because it is not very important and its only effect is a renormalization of the single-particle energies. These are, in fact, influenced far more by particle–hole correlations, which are in any case not described in the proper way by the pairing force. In this case we have

$$\left.\begin{matrix} u_k^2 \\ v_k^2 \end{matrix}\right\} = \frac{1}{2}\left[1 \pm \frac{\epsilon_k - \lambda}{\sqrt{(\epsilon_k - \lambda)^2 + \Delta^2}}\right] . \tag{6.59}$$

The $v_k^2 = \langle \mathrm{BCS}|a_k^+ a_k|\mathrm{BCS}\rangle$ are the *occupation probabilities* for the different single-particle states.

Again, from (6.59) we see that in the limit $G \to 0$, that is, $\Delta \to 0$, the $v_k^2 = 1$ for occupied levels and $v_k^2 = 0$ for unoccupied ones. In this case, v_k^2 is a step function (Fig. 6.5), whereas in the interacting case ($\Delta \neq 0$) the step function is somewhat smeared out. Due to the interaction, particles are scattered from below to above the Fermi surface. This yields a partial depletion of the states below and a partial filling of the states above the Fermi level

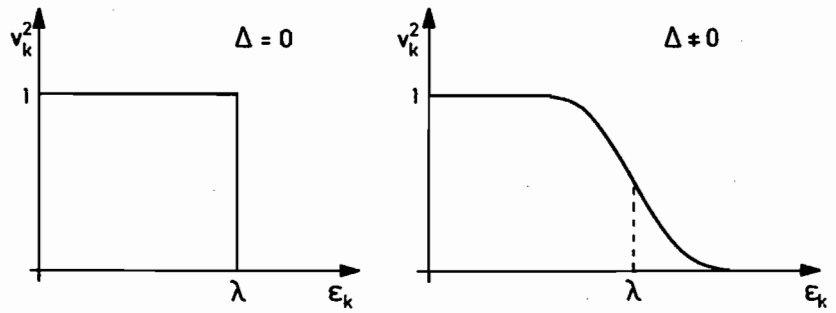


Figure 6.5. The occupation probabilities v_k^2 in the noninteracting case ($\Delta = 0$) and in the interacting case ($\Delta \neq 0$).

(Fig. 6.5). The *gap equation* (6.54) takes the simple form

$$\Delta = \frac{G}{2} \sum_{k>0} \frac{\Delta}{\sqrt{(\epsilon_k - \lambda)^2 + \Delta^2}} . \tag{6.60}$$

In the special case of a single j-shell, all ϵ_k are equal. From (6.58), (6.59), and (6.52) we see that in this case (even taking into account the term $G \cdot v^2$) all v_k^2 are equal. From the particle-number condition (6.39), we find

$$v_k = \sqrt{\frac{N}{2\Omega}} , \qquad u_k = \sqrt{1 - \frac{N}{2\Omega}} . \tag{6.61}$$

The corresponding energy is ($\epsilon_k = 0$):

$$E_{\mathrm{BCS}}^{(N)} = -\frac{1}{2} G \cdot N \cdot \Omega \left(1 - \frac{N}{2\Omega} + \frac{N}{2\Omega^2} \right). \tag{6.62}$$

This expression agrees for large N up to order $1/\Omega$ with the exact formula of Eq. (6.15). Therefore, we see that the BCS ansatz is a good approximation, well suited to treat the nuclear pairing correlations.

The uncertainty in the particle number can also be obtained according to (6.44) and (6.61):

$$\frac{\Delta N}{N} = \frac{1}{\sqrt{N}} \sqrt{2 - \frac{N}{\Omega}} . \tag{6.63}$$

The gap in this model is given using (6.57) and (6.61) as

$$\Delta = G \cdot \sqrt{\frac{N}{2} \left(\Omega - \frac{N}{2} \right)} . \tag{6.64}$$

The gap thus has a parabolic dependence on the number of particles in the shell and is zero for empty or filled shells. For $N = \Omega$, (half-filled shells), we find

$$2\Delta = G \cdot \Omega. \tag{6.65}$$

As we will see later (Sect. 6.3.4), it is not by accident that 2Δ is equal to the excitation energy (6.14) of the first excited state and it will become clear why Δ is called the (energy) gap.

6.3.4 Bogoliubov Quasi-particles—Excited States and Blocking

The main advantage of writing the ground state in the form (6.31) or (6.35) is that despite being very similar to the exact ground state of the seniority model (condensate of pairs), and therefore containing correlations between pairs of particles, $|\mathrm{BCS}\rangle$ can at the same time be written as a product state of a new type of fermions: the *Bogoliubov quasi-particles*. The concept of quasi-particles as a general concept in many-body physics will be discussed in Chapter 7. In fact, it is easily verified that

$$|\mathrm{BCS}\rangle \propto \prod_{k \gtrless 0} \alpha_k |-\rangle \tag{6.66}$$

with

$$\begin{aligned} \alpha_k^+ &= u_k a_k^+ - v_k a_{\bar{k}}, \\ \alpha_{\bar{k}}^+ &= u_k a_{\bar{k}}^+ + v_k a_k, \end{aligned} \tag{6.67}$$

and the following fermion commutation relations hold:

$$\{\alpha_k, \alpha_{k'}\} = 0; \qquad \{\alpha_k, \alpha_{k'}^+\} = \delta_{kk'}, \tag{6.68}$$

where we have used (6.34) and the usual phase convention

$$u_{\bar{k}} = u_k > 0; \qquad v_{\bar{k}} = -v_k < 0; \qquad k > 0. \tag{6.69}$$

The ansatz (6.67) for quasi-particles is very similar to the ansatz for HF-quasi-particles in the Lipkin model (5.50) and we could therefore have used the same techniques for their solution. In the latter case, we can show that real coefficients ($\varphi = 0$) give the lowest energy. The same is also possible here if the force has only matrix elements $\bar{v}_{k\bar{k}k'\bar{k}'} < 0$. Since this condition is very reasonable, we shall use only real coefficients u_k and v_k.

From Eqs. (6.67) and (6.52) we see that a quasi-particle has some properties of a bare particle and some of a bare hole: Above the Fermi surface (v_k^2 small) it is nearly a particle, while below the Fermi surface (u_k^2 small) it is nearly a hole.

We see from Eq. (6.67) that by using this very useful trick of a linear "*Bogoliubov*" *transformation* we have achieved a representation of the ground state of pairwise *interacting* particles in terms of a gas of *non-interacting* quasi-particles. This is, of course, in many practical cases a very helpful feature. The price we have to pay is that the transformation (6.67) clearly does not conserve particle number because we mix creation and annihilation operators. If we assume that not only the ground state is well

represented by a product state of quasi-particles, but also *excited states*, then the Hamiltonian H_{qp}, which corresponds to this gas of non-interacting gas of quasi-particles, is given by

$$H_{qp} = \langle \mathrm{BCS}|H'|\mathrm{BCS}\rangle + \sum_{k \gtrless 0} E_k \alpha_k^+ \alpha_k, \tag{6.70}$$

where the constant $\langle H' \rangle$ takes account of the fact that we have

$$\alpha_k|\mathrm{BCS}\rangle = 0 \qquad \text{for all} \quad k \gtrless 0, \tag{6.71}$$

therefore H_{qp} has the right ground state expectation value. The quasi-particle energies E_k are a straightforward generalization of the definition of those for real particles (for details of the calculations see Chap. 7)

$$\begin{aligned} E_k &= \langle \mathrm{BCS}|\alpha_k H' \alpha_k^+|\mathrm{BCS}\rangle - \langle \mathrm{BCS}|H'|\mathrm{BCS}\rangle \\ &= \sqrt{\tilde{\epsilon}_k^2 + \Delta_k^2}\ . \end{aligned} \tag{6.72}$$

The one-quasi-particle states

$$\alpha_{k_1}^+|\mathrm{BCS}\rangle = a_{k_1}^+ \prod_{k \neq k_1} (u_k + v_k a_k^+ a_{\bar{k}}^+)|-\rangle, \tag{6.73}$$

$$\alpha_{\bar{k}_1}^+|\mathrm{BCS}\rangle = a_{\bar{k}_1}^+ \prod_{k \neq k_1} (u_k + v_k a_k^+ a_{\bar{k}}^+)|-\rangle \tag{6.74}$$

obviously have the energy $\langle H' \rangle + E_k$. They are a superposition of states with odd particle number and describe a nucleus with an odd number of nucleons. According to the quantum number k_1 this state can be either the ground state or an excited state.

The two-quasi-particle states

$$\begin{aligned} &\alpha_{k_1}^+ \alpha_{k_2}^+|\mathrm{BCS}\rangle = a_{k_1}^+ a_{k_2}^+ \prod_{k \neq k_1,\, k_2} (u_k + v_k a_k^+ a_{\bar{k}}^+)|-\rangle, \qquad k_2 \neq \bar{k}_1, \\ &\alpha_{k_1}^+ \alpha_{\bar{k}_1}^+|\mathrm{BCS}\rangle = (-v_{k_1} + u_{k_1} a_{k_1}^+ a_{\bar{k}_1}^+) \prod_{k \neq k_1} (u_k + v_k a_k^+ a_{\bar{k}}^+)|-\rangle, \qquad k_2 = \bar{k}_1, \end{aligned} \tag{6.75}$$

have the energy $\langle H' \rangle + E_{k_1} + E_{k_2}$. They describe excited states in the even system. In this case one pair is broken and the excitation energy is

$$E_{k_1} + E_{k_2} \geqslant 2\Delta. \tag{6.76}$$

The first excited state in the even system thus lies at least 2Δ higher than the ground state.

In the special case of a single j-shell, the excitation energy is given by

$$E_k + E_{k'} = 2\sqrt{\tilde{\epsilon}^2 + \Delta^2}\ . \tag{6.77}$$

With (6.58), (6.41), and (6.62) we find ($\epsilon_k = 0$)

$$\tilde{\epsilon} = G\frac{\Omega - N}{2}, \tag{6.78}$$

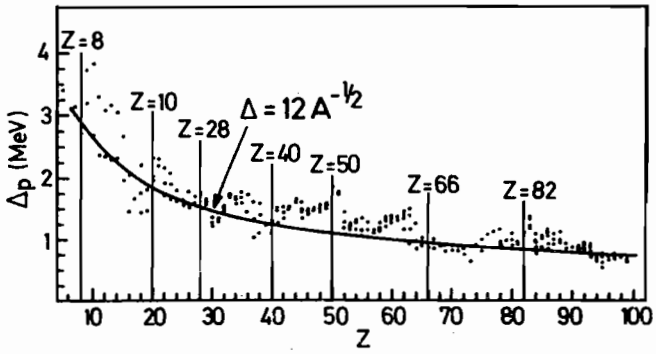


Figure 6.6. Dependence of the proton gap on the proton number. (From [BM 69].)

which, together with (6.64), gives

$$E_k + E_{k'} = G \cdot \Omega, \tag{6.79}$$

in accordance with the result (6.14) of the seniority model. From (6.78) we see that only in the middle of the shell is the excitation energy 2Δ.

In the odd system with the ground state k_0 ($\tilde{\epsilon}_{k_0} \simeq 0$), the excitation energy is

$$E_k - E_{k_0} = \sqrt{\tilde{\epsilon}_k^2 + \Delta^2} - \Delta. \tag{6.80}$$

For small excitation energy ($\tilde{\epsilon}_k \ll \Delta$) we therefore find a high level density in these odd systems.

We also can explain the odd–even mass difference by the following consideration: The ground state energies E_N^{GS} are given for any N by

$$E_{N+2}^{GS} = E_N^{GS} + 2\lambda; \qquad E_{N+1}^{GS} = E_N^{GS} + \lambda + E_{k_0},$$

and therefore the odd–even mass difference is given by:

$$\frac{1}{2}\left\{E_{N+2}^{GS} - E_{N+1}^{GS} - \left(E_{N+1}^{GS} - E_N^{GS}\right)\right\} = -E_{k_0} \simeq -\Delta. \tag{6.81}$$

Equation (6.81) is often exploited to determine the gap empirically from the measured binding energies. It has been found that on the average the gap follows the relation $\Delta = 12 \cdot A^{-1/2}$ as a function of nucleon number A [NA 62, ZGS 67] (Fig. 6.6).

The above considerations give a qualitative understanding of the structure of many states in superfluid nuclei. There remain, however, a few important points to take into account in a more detailed investigation:

(i) The Chemical Potential. As we have seen in (6.53), the chemical potential λ is determined in such a way that the average particle number in the BCS ground state has the correct value. With the same λ, we find for a

one-quasi-particle state $|k\rangle = \alpha_k^+ |\mathrm{BCS}\rangle$

$$\langle k|\hat{N}|k\rangle = N + u_k^2 - v_k^2. \tag{6.82}$$

This is $N \pm 1$ only for levels k which are far away from the Fermi surface. For the levels in the vicinity of the Fermi surface the average particle number is wrong. Since the energy depends strongly on the average particle number, one should readjust the chemical potential λ for the different levels in odd nuclei and also for the excited levels in even nuclei. As long as we have not done this, we should use the operator $H' = H - \lambda\hat{N}$ instead of H for the calculation of excitation energies (as we have done so far). This can, for instance, be seen if we correct for the wrong particle number in the state $|k\rangle$ of Eq. (6.82):

$$\begin{aligned} E_{N+1}^k &= \langle k|H|k\rangle + \frac{dE}{dN}(N+1-\langle k|\hat{N}|k\rangle) \\ &= \langle k|H-\lambda\hat{N}|k\rangle + \lambda(N+1) \\ &= E_N^{GS} + \lambda + E_k. \end{aligned}$$

(ii) The Blocking Effect. The occupation probabilities v_k^2 of the BCS ground state [Eq. (6.52)] were determined by the variational principle. The ground state of an odd system is described by the wave function

$$\alpha_{k_1}^+ |\mathrm{BCS}\rangle = a_{k_1}^+ \prod_{k \neq k_1} (u_k + v_k a_k^+ a_{\bar{k}}^+)|-\rangle. \tag{6.83}$$

The unpaired particle sits in the level k_1 and blocks this level. The Pauli principle prevents this level from participating in the scattering process of nucleons caused by the pairing correlations. The level k_1 always stays occupied and the level $\bar{k}$ always stays empty. Only for $k \neq k_1$ do we have $v_k^2 = v_{\bar{k}}^2$ (Fig. 6.7). Using the blocked wave function (6.83) as a trial wave function in the variational principle, we find the same equations for v_k^2 as before. The only difference is that in the calculation of the gap one level is

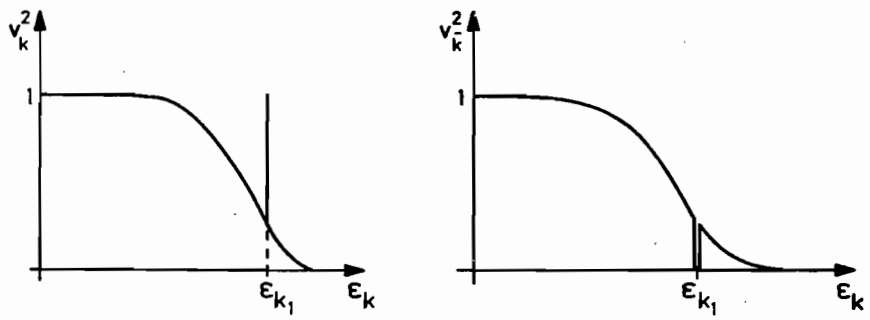


Figure 6.7. Blocking of the state k_1.

"blocked":

$$\Delta = G \sum_{k \neq k_1} u_k v_k. \tag{6.84}$$

The level k_1 has to be excluded from the sum because it cannot contribute to the pairing energy. The chemical potential is determined by

$$N = 1 + 2 \sum_{k \neq k_1} v_k^2. \tag{6.85}$$

Similar equations hold for the case of a higher number of blocked levels, as in the case of two-quasi-particle states.

The change in Δ and in the u_ks and v_ks is called the blocking effect. These blocking correlations are of the order Ω^{-1} and can often be neglected. However, as pointed out by Soloviev [So 61, Wa 62b] and Nilsson and Prior [NP 61], the correction may be large in some cases. This can happen particularly for deformed nuclei where, although there may be 20 levels in the spectrum, only 4 or 5 contribute appreciably to the sum of Eq. (6.50). Clearly the blocking of one or two such levels in such a case has a big effect so that one cannot simply equate the excitation energies to quasi-particle energies, but must evaluate the total energy of the excited state and subtract it from the vacuum energy. For two-quasi-particle states the corrected energy ($\simeq$1.4 MeV) is always smaller than the quasi-particle energy ($\simeq$1.7 MeV), but much larger than the free-particle energy ($\simeq$0.15 MeV). One problem that arises in this corrected theory, in which the v_k can be appreciably different in the ground state and the excited states, is that these states are no longer automatically orthogonal, although sometimes a different quantum number, such as the spin or the parity, guarantees the orthogonality. In particular, two-quasi-particle states of spin 0^+ are not orthogonal to the ground state in such cases.

6.3.5 Discussion of the Gap Equation

For all practical cases, the BCS-equations have to be solved on a computer and the question arises of how many levels one should include for such calculations, that is, we have to investigate which states contribute most to the sum appearing in the gap equation (6.54). For this purpose, let us first consider the gap Δ_k for a state close to the Fermi level. In this case the main contributions to the sum come from states in the vicinity of the Fermi level. The reason for this is that (i) in this case the expression $\Delta_{k'}/(\tilde{\epsilon}_{k'}^2 + \Delta_{k'}^2)^{1/2} \simeq 1$ ($\tilde{\epsilon}_{k'} \simeq 0$), and (ii) the matrix elements $v_{k\bar{k}k'\bar{k}'}$ for $k' \simeq k$ (strongly overlapping wave functions) are larger than the other matrix elements.

Conversely, if k is far from the Fermi surface, all the terms in the sum of Eq. (6.54) are small. For $k' \simeq k$, the overlap of the wave functions is still large, but the factor $\Delta_{k'}/(\tilde{\epsilon}_{k'}^2 + \Delta_{k'}^2)^{1/2}$ is now small because of $\tilde{\epsilon}_k^2 \gg \Delta_k^2$. For

states k' at the Fermi surface it is just the other way around, since there the matrix elements are very small (k and k' usually belonging to different principal quantum numbers).

The effect of the pairing force is therefore restricted to the neighborhood of the Fermi surface, that is, only there is Δ_k different from zero. We may now understand why it might be a valid approximation to take a constant pairing force in the vicinity of the Fermi surface and have zero matrix elements elsewhere. On the other hand, if we had taken a constant matrix element over the whole configuration space, the gap would have diverged.

For spherical, only weakly deformed nuclei the single-particle levels exhibit a pronounced shell structure. The main contribution to the sum in the gap equation comes from transitions within the same shell. We can therefore more or less treat each shell separately. From our discussion above we expect for inner shells $\Delta_k \simeq 0$. The pairing force will therefore only be effective in the partially filled shell, which is often called the Λ-shell. This leads back to the seniority model. For strongly deformed nuclei the shell structure is completely washed out, nevertheless, the sum only runs over a group of levels in the neighborhood of the Fermi surface, also called the Λ-shell.

The restriction of the pairing to the vicinity of the Fermi level is also the reason that neutrons and protons can be treated separately (at least for heavy nuclei). For nuclei with $A \sim 150$, the neutron excess is $(N-Z) \geqslant 20$. The neutron and proton levels close to the Fermi energy, therefore, have very small overlap compared to that of protons or neutrons alone. Therefore, neglecting the proton–neutron pairing, the total wave function may be represented as a product of the proton and neutron functions:

$$|\mathrm{BCS}\rangle = \prod_{k_1>0} \left(u_{k_1}^{(p)} + v_{k_1}^{(p)} a_{k_1}^{(p)+} a_{\bar{k}_1}^{(p)+}\right) \prod_{k_2>0} \left(u_{k_2}^{(n)} + v_{k_2}^{(n)} a_{k_2}^{(n)+} a_{\bar{k}_2}^{(n)+}\right) |-\rangle. \tag{6.86}$$

In spite of these simplifying assumptions, the BCS equations can only be solved numerically. In order to study the influence of the interaction between pairs of nucleons in a qualitative manner we restrict ourselves to a pure pairing force (6.55) within the Λ-shell. In the following considerations, we therefore neglect the other shells. The gap equation is then of the form

$$\Delta = \frac{1}{2} G \sum_{k>0}^{\Lambda} \frac{\Delta}{\sqrt{\tilde{\epsilon}_k^2 + \Delta^2}}, \tag{6.87}$$

where the sum runs over the Λ-shell only. Equation (6.87) always has the trivial solution $\Delta = 0$, that is, $u_k \cdot v_k = 0$, which for no or sufficiently weak pairing force is the only solution. However, for

$$\frac{1}{2} G \sum_{k>0}^{\Lambda} \frac{1}{|\tilde{\epsilon}_k|} > 1 \tag{6.88}$$

there exists a second nontrivial solution $\Delta > 0$. This always happens if the pairing force is sufficiently strong or if the Λ-shell is sufficiently large. However, this sharp transition is a consequence of particle number violation and it is somewhat smeared out in a more extended theory [MPR 65]. For an infinite system, particle number violation is negligible, and the transition is always sharp.

On the other hand, (6.87) can also be used to determine the strength G of the pairing force since, and, as we have already said, Δ can be determined empirically from the odd–even effect. It still, however, depends on the "cutoff" Λ.

6.3.6 Schematic Solution of the Gap Equation

It is very instructive to have an *analytic solution* of the gap equation [Be 59]. This can only be achieved in a very schematic model. We image a well deformed nucleus for which the level density is almost uniform and the average level spacing $\Delta\epsilon$ is very small compared with Δ (for deformed nuclei in the rare earth region $\Delta \simeq 1$ MeV and $\Delta\epsilon \simeq 0.1$–$0.2$ MeV). We can then replace the sum in Eq. (6.87) approximatively by an integration over the Λ shell: $\epsilon' < \epsilon < \epsilon''$; $a = \epsilon' - \lambda$; $b = \epsilon'' - \lambda$,

$$1 = \frac{1}{2} G \int_a^b \frac{1}{\sqrt{\epsilon^2 + \Delta^2}} \rho(\epsilon)\, d\epsilon, \tag{6.89}$$

where $\rho(\epsilon)$ is the level density. We furthermore assume that $\rho(\epsilon)$ is approximately constant within the Λ-Shell ($\rho \simeq \bar{\rho}$). We introduce the dimensionless quantity

$$\eta = \frac{1}{\bar{\rho} G} \tag{6.90}$$

and obtain after integration

$$2\eta = \operatorname{arsinh} \frac{b}{\Delta} - \operatorname{arsinh} \frac{a}{\Delta}. \tag{6.91}$$

Solving for Δ yields

$$\Delta = \frac{1}{\sinh 2\eta} \sqrt{b^2 + a^2 - 2ab \cosh 2\eta}\ . \tag{6.92}$$

The average particle number in the BCS model is given by Eq. (6.39) and determines the chemical potential λ. The number of particles in the Λ shell is given by (again we replace the sum in (6.39) by an integral):

$$N = 2 \int_a^b \frac{1}{2} \left(1 - \frac{\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \right) \bar{\rho}\, d\epsilon. \tag{6.93}$$

The number of levels in the Λ-shell is given by

$$\Omega = \bar{\rho}(b - a). \tag{6.94}$$

Introducing the occupation factor

$$\chi_N = 1 - \frac{N}{\Omega}, \tag{6.95}$$

we get, after a lengthy but straightforward calculation using (6.92),

$$\lambda = \frac{\epsilon' + \epsilon''}{2} - \frac{\epsilon'' - \epsilon'}{2}\chi_N \cdot \operatorname{ctgh}\eta. \tag{6.96}$$

Inserting (6.96) in (6.92), we get:

$$\Delta = \frac{\epsilon'' - \epsilon'}{2\sinh\eta}\sqrt{1-\chi_N^2}\ . \tag{6.97}$$

In the limit of weak coupling, that is, for $G\cdot\bar{\rho} \ll 1$, we find:

$$\Delta \propto e^{-(1/\rho G)}. \tag{6.98}$$

We therefore have from (6.98) the important result that the gap cannot be developed as a power series in the interaction strength.

Equation (6.97) has been derived under the condition that Δ is not smaller than the level distance, that is,

$$\Delta\cdot\bar{\rho} > 1 \tag{6.99}$$

or

$$\frac{1}{2\sinh\eta}\sqrt{N(2\Omega - N)} > 1. \tag{6.100}$$

For $\eta \leqslant 1$ (strong pairing correlations), this inequality is always fulfilled. For $\eta \gg 1$, Eq. (6.100) is only satisfied for large enough values of $N(2\Omega - N)$.

The ground state energy in the BCS-theory is given by (6.56). Its contribution to the ground state expectation value of H' is

$$\langle \mathrm{BCS}|H'|\mathrm{BCS}\rangle = 2\int_a^b \frac{1}{2}\left(1 - \frac{\epsilon}{\sqrt{\epsilon^2+\Delta^2}}\right)\epsilon\bar{\rho}\,d\epsilon - \frac{\Delta^2}{G}. \tag{6.101}$$

Since H' differs from H by $\lambda\cdot N$, using (6.96) and performing the integral in (6.101) we get for the ground state energy:

$$E_{\mathrm{BCS}} = \langle \mathrm{BCS}|H|\mathrm{BCS}\rangle = \tfrac{1}{2}(\epsilon'' + \epsilon')N - \tfrac{1}{4}\Omega(\epsilon'' - \epsilon')(1-\chi_N^2)\operatorname{ctgh}\eta. \tag{6.102}$$

We are interested in that part of the ground state energy which comes from the pairing force alone. For this purpose, we first calculate the energy for $G = 0$ which corresponds to the limit $\eta \to \infty$:

$$E^0 = \tfrac{1}{2}(\epsilon' + \epsilon'')\cdot N - \tfrac{1}{4}\Omega(\epsilon'' - \epsilon')(1-\chi_N^2). \tag{6.103}$$

The contribution E^P of the pairing force to the ground state energy is therefore given by:

$$E^P = E_{\mathrm{BCS}} - E^0. \tag{6.104}$$

In the case of a half-filled shell where $N = \Omega$, we find, using Eqs. (6.102),

(6.103), (6.94), and (6.97),

$$E^P = -\frac{1}{4}\Omega(\epsilon'' - \epsilon)(1-\chi_N^2)(\mathrm{ctgh}\,\eta - 1) = -\frac{1}{2}\Delta^2\bar{\rho}(1-e^{-2\eta}). \quad (6.105)$$

This is the *pairing energy* for neutrons or protons. For heavy nuclei it is possible to estimate G and Δ from the odd–even effect. Nilsson and Prior [NP 61] found for protons $G_P \simeq 17/A$ MeV and for neutrons $G_n \simeq 25/A$ MeV. The energy gap Δ is roughly 1 MeV. Assuming a characteristic level density $1/\bar{\rho} \simeq 0.5$ MeV, we get $\eta \simeq A/100$. In this case we can neglect the term $e^{-2\eta}$ in (6.105) and obtain for the total pairing energy, which is more or less the same for protons and neutrons,

$$E^P_{\text{total}} = E^P_{(p)} + E^P_{(n)} \simeq -\bar{\rho}\Delta^2 \simeq -2[\mathrm{MeV}]. \quad (6.106)$$

The total pairing energy is therefore extremely small compared to the total binding energy. This is a general feature of all kinds of correlation energies, which does not mean, however, that they cannot play a decisive role in the explanation of many problems.

As we have seen, the BCS wave function represents a state which is not an eigenstate of the particle number operator. The whole theory is only valid if $|\mathrm{BCS}\rangle$ is well localized around the actual particle number. In order to get an estimate, we therefore calculate the *mean square deviation of the particle number* (6.44):

$$(\Delta N)^2 = \sum_{k>0} \frac{\Delta_k^2}{\tilde{\epsilon}_k^2 + \Delta_k^2}. \quad (6.107)$$

Within the schematic model it follows that

$$(\Delta N)^2 = \bar{\rho}\Delta\left(\arctan\frac{b}{\Delta} - \arctan\frac{a}{\Delta}\right). \quad (6.108)$$

In the limit of weak pairing correlations or small level density we get

$$(\Delta N)^2 = \bar{\rho}\cdot\Delta\cdot\pi. \quad (6.109)$$

This approximation is roughly fulfilled for strongly deformed nuclei. Taking characteristic values for $\bar{\rho}$ and G, we obtain $\eta \simeq 4,\ldots,5$ and for the mean deviation $\Delta N \simeq \sqrt{6}$.

In the other limit of strong pairing ($\eta \ll 1$), we take only the leading terms of (6.108) with respect to η:

$$(\Delta N)^2 = \Omega(1-\chi_N^2) = 2N\left(1 - \frac{N}{2\Omega}\right). \quad (6.110)$$

Consequently, for a half-filled shell ($N=\Omega$) one finds $\Delta N = \sqrt{N}$, whereas for only one pair ($N=2$) we obtain $\Delta N \simeq 2$. A typical example for a half-filled shell with $\eta \ll 1$ is ^{202}Pb. Its BCS wave function contains large amounts of ^{204}Pb and ^{200}Pb and also of ^{206}Pb and ^{198}Pb. We have to keep in mind, however, that the Fermi surfaces λ and the gap Δ are quite the same for all these nuclei, and therefore the uncertainty in the particle number is not as bad as it appears.

As we have seen, the very simple BCS model allows us to explain many features in nuclei arising from the short-range correlations of nucleons. We have only treated the pairing of neutrons and protons separately, neglecting the neutron–proton interaction. For heavy nuclei this was valid because the proton and neutron wave functions usually belong to different principal quantum numbers and the pairing matrix elements are small. For light nuclei (s–d shell!), however, this is no longer true, and we also have to include neutron–proton pairing. Before we mention very briefly some of those calculations, we want to reconsider Hartree–Fock theory and deformations and pairing from the more general Hartree–Fock–Bogoliubov theory which, as we will see, embraces all the models we have treated so far.